Software Engineering

Sample Questions

1. Define Software Engineering. A software product has been delivered late and exceeds the budget. Analyze possible reasons for this failure and suggest how software engineering practices could have prevented it. [5]
2. Describe a process framework in your own words. When we say that framework activities are applicable to all projects, does this mean that the same work tasks are applied for all projects, regardless of size and complexity? Explain [5]
3. Explain the phases of the Waterfall Model with a neat diagram. Suppose you are developing a *banking application* where requirements are clearly defined and stable. Analyze why the Waterfall Model would be a suitable choice for this project. [5]
4. What is V-model in Software development? A *medical software system* must undergo strict testing and validation for patient safety. Justify why the V-Model is more effective than the Waterfall Model in this scenario. [5]
5. Your client needs a *university management system* where some core features must be delivered quickly, and additional features can be added later. Analyze which Process model can be practiced here and how the model would be used to manage this project effectively. [5]

A bookstore plans to develop a Book Store Management System (BSMS) to streamline both in store and online operations. The system will allow customers to browse books, search by title, author, or category, view detailed information, add selected books to a shopping cart, place orders, make online payments, and track their order status from placement to delivery. Customers will also be able to submit reviews and feedback on purchased books. Store employees will use the system to manage daily tasks such as adding new books to the catalog, updating stock quantities, processing customer orders by confirming, packing, and shipping them, handling complaints, and managing returns or refunds. The system will maintain several key data stores, including customer profiles, book catalog details, supplier information, order histories, payment records, and inventory levels. The BSMS is expected to efficiently support workflows such as order processing, inventory restocking, supplier management, and return or refund operations, ensuring smooth interaction between customers, employees, and administrators.

1. a) Identify any five major use cases for the BSMS from the perspective of different actors. [10]
   b) Draw a Use Case Diagram showing the actors and their interactions with the system.
2. Prepare class-based modeling for the Book Store Management System by identifying potential classes using proper selection criteria, creating class cards with attributes and operations, and drawing the corresponding class diagram. [10]
3. What is a Data Flow Diagram (DFD)? Explain its purpose in requirements modeling and draw DFD up to Level-2 for the BSMS showing major processes, data stores, and external entities. [10]
4. a) Develop a detailed Sequence Diagram for the BSMS that depicts the interaction among the Customer, System, Payment Service, and Store Employee during the complete [10]

online book ordering process, covering steps such as book selection, adding to cart, order placement, payment verification, order confirmation, and notification.

b) Prepare a comprehensive State Transition Diagram for the Order Lifecycle in the BSMS, illustrating all major states and the transitions triggered by user actions or system events.